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Probe card and manufacturing method thereof

Abstract

The disclosure provides a probe card comprising a circuit substrate, a space transformer, and a first coaxial cable. The circuit substrate has a first surface and a second surface, wherein a through hole penetrates the circuit substrate from the first surface to the second surface. The space transformer is arranged on the second surface corresponding to the through hole and has a probe setting surface opposite to the second surface. The first coaxial cable has a first conducting part and a first insulating part, wherein the first coaxial cable is inserted into the through hole, and a first end surface of the first conducting part is flush with the probe setting surface.

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Background/Summary

FIELD OF THE INVENTION
(1) The disclosure relates to a probe card and manufacturing method thereof, in particular, a probe card and manufacturing method for reducing impedance mismatching.
BACKGROUND OF THE INVENTION
(2) In the semiconductor industry, probe cards are usually used for checking the quality or function of products. Therefore, the efficiency and accuracy of the probe card while measuring are key factors for the yield rate of products. For instance, signal wires of a probe card may affect the test result (e.g., parameters such as bandwidth), especially for semiconductor products needed to be tested with high frequency signal (e.g., communication products).
(3) Accordingly, how the impedance matching/continuity of the test signal wire or the ability to shield noise can be improved are issues for probe cards to overcome when performing high-frequency signal testing.
SUMMARY OF THE INVENTION
(4) One of the purposes of the disclosure is to improve the impedance matching/continuity of the signal wire of the probe card while measuring, especially for high-frequency signal tests.
(5) One of the purposes of the disclosure is to improve the probe card's ability to resist noise while

measuring, especially for high-frequency signal tests.
(6) The disclosure provides a probe card comprising a circuit substrate, a space transformer, and a first coaxial cable. The circuit substrate has a first surface and a second surface, wherein a through hole penetrates the circuit substrate from the first surface to the second surface. The space transformer is arranged on the second surface corresponding to the through hole and has a probe setting surface opposite to the second surface. The first coaxial cable has a first conducting part and a first insulating part, wherein the first coaxial cable is inserted into the through hole, and a first end surface of the first conducting part is flush with the probe setting surface.
(7) The disclosure provides a method for manufacturing a probe card. The method comprises: forming a through hole on a circuit substrate; arranging a space transformer on a second surface of the circuit substrate at a position corresponding to the through hole; inserting a coaxial cable into the through hole from outside a first surface of the circuit substrate, wherein a portion of the conducting part of the coaxial cable extends out from a probe setting surface of the space transformer; and removing, by a machining method, the portion of the conducting part of the coaxial cable extending out from the probe setting surface to make the end surface of the conducting part flush with the probe setting surface.
(8) In summary, the conducting part of the coaxial cable is directly inserted into the through hole without other wire connecting, and the conducting part of the coaxial cable is flush with the probe setting surface. In other words, the impedance of the signal path will be uniform to improve the impedance matching/continuity of the signal path and the ability to resist noise.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIGS. **1**A-**1**B are schematics of the cross section of the probe card according to an embodiment of the disclosure.
(2) FIGS. **2**A-**2**B are schematics of the arranging path of the coaxial cable according to an embodiment of the disclosure.
(3) FIG. **3** is a schematic of disposing the filler according to an embodiment of the disclosure.
(4) FIG. **4**A is a schematic of arranging a plurality of coaxial cables into the probe card according to an embodiment of the disclosure.
(5) FIG. **4**B is a schematic of the distance between two end surfaces of the plurality of coaxial cables according to an embodiment of the disclosure.
(6) FIGS. **5**A-**5**C are schematics of arranging probe head to the probe card according to an embodiment of the disclosure.
(7) FIGS. **6**A-**6**B are schematics of coupling the end surface of the coaxial cable with the probe contact pad according to an embodiment of the disclosure.
(8) FIG. **7** is a flowchart of the method for manufacturing the probe card according to an embodiment of the disclosure.
(9) The accompanying drawings are presented to aid in the description of various aspects of the disclosure and are provided solely for illustration of the aspects. In order to simplify the drawings and highlight the contents to be presented in the drawings, the well-known structures or elements in the drawings may be drawn in a simple schematic manner or presented in an omitted manner. For example, the number of elements may be singular or plural.
(10) These drawings are provided only to explain these aspects and not to limit thereof.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS
(11) It should be understood that, even though the terms such as “first”, “second”, “third” may be used to describe an element, a part, a region, a layer and/or a portion in the present specification, but these elements, parts, regions, layers and/or portions are not limited by such terms. Such terms

are merely used to differentiate an element, a part, a region, a layer and/or a portion from another element, part, region, layer and/or portion. Therefore, in the following discussions, a first element, portion, region, portion may be called a second element, portion, region, layer or portion, and do not depart from the teaching of the present disclosure. The terms “comprise”, “include” or “have” used in the present specification are open-ended terms and mean to “include, but not limit to.”

(12) As used herein, the term “coupled to” in the various tenses of the verb “couple” may mean that element A is directly connected to element B or that other elements may be connected between elements A and B (i.e., that element A is indirectly connected with element B).

(13) The terms “about”, “approximate” or “essentially” used in the present specification include the value itself and the average values within the acceptable range of deviation of the specific values confirmed by a person having ordinary skill in the present art, considering the specific measurement discussed and the number of errors related to such measurement (that is, the limitation of the measurement system). For example, “about” may mean within one or more standard deviations of the value itself, or within ±30%, ±20%, ±10%, ±5%. In addition, “about”, “approximate” or “essentially” used in the present specification may select a more acceptable range of deviation or standard deviation based on optical property, etching property or other properties. One cannot apply one standard deviation to all properties.

(14) Refer to FIGS. **1**A-**1**B. FIGS. **1**A-**1**B illustrate the probe card **100** comprising the circuit substrate **110**, the space transformer **120** and the first coaxial cable **130**. The circuit substrate **110** has the first surface **111** and the second surface **112**, wherein the through hole **114** penetrates the first surface **111** and the second surface **112**. The space transformer **120** is arranged on the second surface **112** corresponding to the through hole **114** and has a probe setting surface **121** opposite to the second surface **112**. The first coaxial cable **130** has the first conducting part **131** and the first insulating part **132**. The first coaxial cable **130** is inserted into the through hole **114**, and the first end surface **1311** of the first conducting part **131** is flush with the probe setting surface **121**.

(15) More specifically, the circuit substrate **110** can be a printed circuit board which can be stiff or flexible. The circuit substrate **110** can be a multi-layer structure, and each layer of the multi-layer structure can be arranged as a trace, insulation layer or grounding layer. The circuit substrate **110** of the disclosure is not limited to printed circuit boards. For example, the material of the circuit substrate **110** can be FR4, glass (e.g., ITO substrate) or semiconductor (e.g., silicon substrate).

(16) The space transformer **120** is configured to integrate loose signal wires or conducting wires to provide a better connection between the probe setting surface **121** of the space transformer **120** and the probe head or object to be tested, such as a die or chip. In an embodiment, one or more through hole may be preset on the probe setting surface **121** of the space transformer **120** to provide arranging position(s) for, for example but not limit to, the first coaxial cable **130** or other conducting wire **140**. It should be noted that, when there is a need to transmit high frequency signals, coaxial cables with better shielding performance are used. If the coaxial cables are used as grounding wire(s) or wire(s) for transmitting low-frequency signals, enameled wires or thin wires may be selected for space or cost considerations.

(17) The cross-section of the first conducting part **131** and the first insulating part **132** of the first coaxial cable **130** show concentric circles. In other words, the first conducting part **131** is covered by the first insulating part **132** to provide a shielding effect and/or an insulating effect. It should be noted that, in some case, at least a part of the first insulating part **132** of the first coaxial cable **130** may be removed to expose the first conducting part **131**. In an embodiment, the end surface **1321** of the first insulating part **132** can align with the probe setting surface **121** or extend into the space transformer **120**. In an embodiment, the first insulating part **132** may be removed to avoid being inserted into the through hole **114** or passing through the through hole **114** to reach the space transformer **120**. According to the aforementioned embodiment(s), the first conducting part **131** can be fully or partially covered by the first insulating part **132**. The length of the part of the first insulating part **132** covering the first conducting part **131** can be determined according to, for

example, the space available in the probe card for arranging wire, the tension of the coaxial cable, or the requirements for shielding noise. It should be noted that that the first coaxial cable **130** may comprise a plurality of conducting parts and/or insulating parts with concentric circles cross-sections. For example, a first coaxial cable **130** may comprise a first conducting part used for transmitting signal, and a second conducting part used for grounding or providing bias voltage. In the example, the first coaxial cable **130** may further comprise a first insulating part used to isolate the first conducting part and the second conducting part, and a second insulating part located in the outermost layer of the first coaxial cable **130** and used as shield. However, the number of layers or structure of the coaxial cable in the disclosure are not limited to the aforementioned example or embodiment(s).
(18) In an embodiment, as shown in FIGS. **1**A-**1**B, the space transformer **120** further has an arranging component **124** is arranged along the wall **1141** of the through hole **114**. The arranging component **124** includes a stop part **1241** with an outer diameter larger than the through hole **114**. More specifically, the material of the space transformer **120** can be, but not limited to, plastic or materials with stiffness. The arranging component **124** can be arranged into the through hole **114** from outside the first surface **111**. For example, the arranging component **124** can be placed into the through hole **114** with the stop part **1241** stopped by the first surface **111** of the circuit substrate **110** to prevent the arranging component **124** from sliding or moving out of the through hole **114**. However, the stop part **1241** is not required to fix the arranging component **124** in the through hole **114**. For example, the arranging component **124** can be tightly fitted to the wall **1141** of the through hole **114** so that the friction between the arranging component **124** and the wall **1141** of the through hole **114** can fix the arranging component **124**. In another example, the arranging component **124** may be fixed by fixing elements (e.g., screw) or fixing mechanisms (e.g., tenon). However, in some cases, the arranging component **124** may be one body with the substrate of the probe card. It should be noted that the methods for fixing the arranging component **124** are not limited to the aforementioned examples or embodiments.
(19) In an embodiment, referring to FIGS. **2**A-**2**B, FIG. **2**A illustrates the first coaxial cable **130** inserted into the through hole **114** along the gap G between the arranging component **124** and the through hole **114**. More specifically, the first coaxial cable **130** can be inserted into a preset trench or gap G between the arranging component **124** and the wall **1141** of the through hole **114** to extend the cable to the probe setting surface **121** of the space transformer **120**. In an embodiment, as shown in FIG. **2**B, a through hole **116** can be formed for passing and arranging the first coaxial cable **130**. Therefore, the preset trench, gap G, or the through hole **116** provides a force for fixing or supporting the first coaxial cable **130** to avoid pulling the posterior section **130′** of the first coaxial cable **130** (that is, avoid pulling the part of the first coaxial cable **130** that is not inserted into the through hole **114**) and thus damaging the probe card or needing to rewire the coaxial cable.
(20) In an embodiment, referring to FIG. **3**, FIG. **3** illustrates the filler F disposed in the accommodation space A formed by the space transformer **120** arranged in the through hole **114**. More specifically, after arranging the space transformer **120** on the second surface **112** of the circuit substrate **110** corresponding to the through hole **114**, the accommodation space A is formed by the space transformer **120** and the through hole **114**. After inserting the first coaxial cable **130** or other conducting wires **140** into the through hole **114**, the filler F, such as quick-drying adhesive, cyanoacrylate, epoxy resin or UV curable resin, can be disposed into the accommodation space A. The filler F provides a stable support to avoid pulling the first coaxial cable **130** or other conducting wires **140** and thus damaging the probe card. Furthermore, the material of the filler F can be selected from materials having electrical or magnetic characteristics, such as materials for anti-electromagnetic noise or materials with high voltage tolerance, to provide proper anti-noise effect and improve the signal-to-noise ratio during detecting/measuring.
(21) In an embodiment, referring to FIGS. **4**A-**4**B, FIG. **4**A illustrates the probe card **100** further comprising the second coaxial cable **150** having the second conducting part **151** and the second

insulating part **152**. As shown in FIG. **4**B, the second coaxial cable **150** is inserted into the through hole **114**, and the second end surface **1511** of the second conducting part **151** is flush with the probe setting surface **121**, wherein the spacing L between the first end surface **1311** and the second end surface **1511** is within a range from 150 μm to 200 μm as shown in FIG. **4**B. It should be noted that the spacing L between the first end surface **1311** and the second end surface **1511** is determined, for example, according to the size of the semiconductor product to be tested. More specifically, the diameter of the coaxial cable **130** or the coaxial cable **150** can be in micro meter level or selected according to the size of the semiconductor product to be tested. The diameter of the coaxial cable **130** may be the same as or different from that of the coaxial cable **150** depending on the requirements. Additionally, an exemplary definition of the spacing between the first end surface **1311** and the second end surface **1511** may be the distance L between the center points of the first end surface **1311** and the second end surface **1511**, or, the shortest distance L′ between the edges of the first end surface **1311** and the second end surface **1511**. The definition of spacing L is determined and/or adjusted according to the design rule or requirements.

(22) Before arranging the coaxial cable **130**, the coaxial cable **150**, and/or the conducting wire **140**, the probe setting surface **121** of the space transformer **120** may have preset hole(s). The conducting part of each of the coaxial cable **130**, the coaxial cable **150**, and/or the conducting wire **140** can be inserted into the preset hole(s). The conducting parts **131** and **151** of the coaxial cable **130** and **150** may be treated by a machining method, e.g., cutting or flushing, to remove the parts of the conducting parts **131** and **151** that bulge from the probe setting surface **121** in order to form the first end surface **1311** and the second end surface **1511**. The first end surface **1311** and the second end surface **1511** may also be treated with gold plating, polishing, or other surface treatments to improve flatness, conductivity, or corrosion resistance. However, the disclosure is not limited to any setting method of the coaxial cable **130**, **150**, or the conducting wire **140**.

(23) In an embodiment, referring to FIGS. **5**A-**5**C, FIG. **5**A illustrates the probe card **100** further comprising a probe head **160**. The probe head **160** comprises the probe **161** and the probe rack **162**. The probe rack **162** is arranged on the probe setting surface **121**. The probe **161** is electrically coupled with the first end surface **1311**. More specifically, as shown in FIG. **5**B, the probe rack **162** may include an upper rack **1621** and a lower rack **1622**. However, the probe rack **162** may also be a unibody. A probe path for probe **161** to pass through may be set at the probe rack **162**. The probe **161** is arranged on the probe setting surface **121** by the probe rack **162**. When the spacing between the first end surface **1311**, the second end surface **1511**, and/or the end surface of the conducting wire **140** equals or is similar to the spacing between each of the probes arranged in the probe rack **162**, the probes arranged in the probe rack **162**, e.g., the probe **161**, may be electrically coupled to the first end surface **1311** directly. More specifically, the probe **161** is in direct contact with the first end surface **1311** without any adaptor.

(24) on the other hand, in an embodiment, when the spacing between the first end surface **1311**, the second end surface **1511**, and/or the end surface of the conducting wire **140** does not match the spacing between each of probes arranged in the probe rack **162**, the probe **161** may be electrically coupled with the first end surface **1311** indirectly. More specifically, as shown in FIGS. **6**A-**6**B, the probe setting surface **121** may be configured to set the probe contacting pad(s) **126**, and the first end surface **1311** may be connected to the probe contacting pad **126** via a connecting wire **128**. As shown in FIG. **6**A, it should be noted that the connecting wire **128** may be an additional wire for connecting the probe contacting pad **126** with the first end surface **1311**. Or, as shown in FIG. **6**B, the first conducting part **131** of the first coaxial cable **130** may be used as the connecting wire **128** which directly extends to the probe contacting pad **126**. Through the indirect contact between the probe **161** and the probe contacting pad **126**, the probe **161** is electrically coupled with the first coaxial cable **130** indirectly. After connecting the conducting part **131** of the first coaxial cable **130** to the probe contacting pad **126**, the connecting wire **128** can be treated by a machining method (e.g., cutting or flushing), to make the surface smooth, and subsequently surface processing (e.g.,

coating or plating), but not limited thereto. On the other hand, the connecting wire **128** can be trace line printed or etched on the probe setting surface **121**, or be arranged in one layer of the space transformer **120** made of multilayer circuit boards.

(25) FIG. **7** illustrates the method for manufacturing the probe card comprising: step S**1**, forming a through hole on a circuit substrate; step S**2**, arranging a space transformer on a second surface of the circuit substrate and at a position corresponding to the through hole; step S**3**, inserting a coaxial cable into the through hole from outside a first surface of the circuit substrate, wherein a portion of the conducting part of the coaxial cable extends out from a probe setting surface of the space transformer; and step S**4**, removing, by a machining method, the portion of the conducting part of the coaxial cable extending out from the probe setting surface to make the end surface of the conducting part flush with the probe setting surface.

(26) In an embodiment, after step **3**, the method for manufacturing the probe card further comprises step S**3-1**, disposing a filler in an accommodation space formed between the space transformer and the through hole. It should be noted that the step S**3-1** for disposing filler can be performed at any time after step S**3** according to the production situation. The disclosure does not limit step S**3-1** to the above embodiment.

(27) In an embodiment, the method for manufacturing the probe card further comprises step S**5**, arranging a probe head including a probe and a probe rack, wherein the probe rack is arranged on the probe setting surface, and the probe is electrically connected to the end surface. It should be noted that electrical coupling/connecting can be direct or indirect. Direct coupling refers to direct contact between the probe and the end surface of the conductor, while indirect coupling refers to coupling of the probe and the end surface of the conductor via conducting wires or other active/passive components. The disclosure is not limited to coupling means between the probe and the end surface of the conducting part.

(28) The foregoing disclosure is merely preferred embodiments of the present invention and is not intended to limit the claims of the present invention. Any equivalent technical variation of the description and drawings of the present invention of the present shall be within the scope of the claims of the present invention.

Claims

1. A probe card, comprising: a circuit substrate having a first surface and a second surface, wherein a through hole penetrates the circuit substrate from the first surface to the second surface; a space transformer arranged on the second surface corresponding to the through hole and having a probe setting surface opposite to the second surface; and a first coaxial cable having a first conducting part and a first insulating part, wherein the first coaxial cable is inserted into the through hole, and a first end surface of the first conducting part is flush with the probe setting surface; wherein the space transformer has an arranging component arranged along a wall of the through hole; the arranging component includes a stop part with an outer diameter larger than the through hole; wherein the first coaxial cable is inserted into the through hole along a gap between the arranging component and the wall of the through hole.
2. The probe card of claim 1, wherein a filler is disposed in an accommodation space formed between the space transformer and through hole.
3. The probe card of claim 1, further comprising a second coaxial cable having a second conducting part and a second insulating part, wherein the second coaxial cable is inserted into the through hole, and a second end surface of the second conducting part is flush with the probe setting surface, and wherein a spacing between the first end surface and the second end surface is within a range from 150 μm to 200 μm.
4. The probe card of claim 1, wherein a diameter of the first coaxial cable is in micro meter level.
5. The probe card of claim 1, further comprising a probe head including a probe and a probe rack,

wherein the probe rack is arranged on the probe setting surface, and the probe is electrically connected to the first end surface.

6. A method for manufacturing a probe card, comprising: forming a through hole on a circuit substrate; arranging a space transformer on a second surface of the circuit substrate and at a position corresponding to the through hole, wherein the space transformer has an arranging component arranged along a wall of the through hole; the arranging component includes a stop part with an outer diameter larger than the through hole; inserting a coaxial cable into the through hole from outside of a first surface of the circuit substrate, wherein a portion of a conducting part of the coaxial cable extends out from a probe setting surface of the space transformer, wherein the first coaxial cable is inserted into the through hole along a gap between the arranging component and the wall of the through hole; and removing, by a machining method, the portion of the conducting part of the coaxial cable extending out from the probe setting surface to make the end surface of the conducting part flush with the probe setting surface.

7. The method of claim 6, further comprising: after the coaxial cable is inserted into the through hole, disposing a filler in an accommodation space formed between the space transformer and the through hole.

8. The method of claim 6, further comprising: arranging a probe head including a probe and a probe rack, wherein the probe rack is arranged on the probe setting surface, and the probe is electrically connected to the end surface.